

CSC 380 — Midterm Study Guide (PROVISIONAL · LIGHT · date NV)

Banner — date NEEDS VERIFICATION; scope PROVISIONAL; LIGHT guide.

- **Date:** the syllabus says **"tentatively"** Tue Oct 20, 2026, 5:00–6:15 PM Arizona time, in the regular lecture. The calendar also lists Oct 20. The date stays **Needs verification** until the instructor confirms it.
- **Scope:** coverage is **not published**. It is provisional, from syllabus weeks 1–8: pandas, visualization, probability, random variables and distributions, covariance and correlation, and midterm review.
- **Light only:** this is a concept map plus a small practice set. Do not do dense prep that assumes a confirmed date.

Canonical maps: `csc380-topic-map.md` · `announced-assessments-inventory-2026-09-24.md` · `site-audit-2026-09-24.md`

1. Header

Field	Value	Status
Course	CSC 380 Principles of Data Science (FA26), TuTh 5:00–6:15 PM, M Pacheco ILC 130	Confirmed
Assessment	Midterm (light provisional guide)	Item exists
Date/time	Tentatively Tue Oct 20, 2026, 5:00–6:15 PM Arizona	Needs verification
Format	Not published. Gradescope delivery is not confirmed.	NV
Coverage	Syllabus weeks 1–8	PROVISIONAL
Not yet lectured	Special Distributions (week 6), Covariance/Correlation (week 7), and Midterm Review (week 8) are still ahead	Expected
Guide updated	2026-09-24 (America/Phoenix) — Opus + Content-fold (light)	

All times are **Arizona time (MST)**.

Sources (Content-pass 2026-09-24)

Source	Role
Content Resources/quiz01_03.pdf	Early quiz-style practice grain (descriptive stats; sample space; LOTP+Bayes). Overlaps Quiz 04 published subset.
Content Homework/homework01.zip, homework02.pdf, homework03.pdf	HW practice context (pandas/viz; probability; Bayes/independence). Not midterm solutions.
Syllabus weeks 1-8	PROVISIONAL midterm coverage map

2. Topic checklist (light concept map)

Order	Section ID	Topic (syllabus week — provisional)
1	380-L-intro	Course overview (wk 1)
2	380-L-pandas	Basic data manipulation with pandas (wk 1)
3	380-L-viz	Basic visualization (wk 1)
4	380-L-prob1	Probability intro; conditional probability (wk 2)
5	380-L-comb	Combinations (wk 3) — <i>new in this review</i>
6	380-Q04-indep	Independence (wk 3; also Quiz 04)
7	380-Q04-bayes / 380-Q04-lotp	Bayes and LOTP (Quiz 04 published subset)
8	380-L-rv	Random variables; distributions; discrete vs continuous (wk 4–5)
9	380-L-special	Special distributions (wk 6 — as lectured)
10	380-L-corr	Covariance and correlation (wk 7 — as lectured)
11	380-L-review	Midterm review (wk 8)

Do **not** start the post-midterm material (modeling, regression, classification).

3. Mini explanations

- **pandas.** These are standard pandas; match your class notebooks.

- Filter rows: `df[df["age"] > 30]`
- Select columns: `df[["a", "b"]]`
- Group means: `df.groupby("team")["score"].mean()`
- Missing values: `df.isna().sum()` and `df.dropna()`

- **Chart choice.**

What you want to show	Chart
Distribution of one numeric variable	Histogram or box plot
Two numeric variables	Scatter plot
Comparison across categories	Bar chart
Change over time	Line chart

- **Combinations.** $C(n, k) = n! / (k!(n-k)!)$ counts **unordered** selections. Permutations $P(n, k) = n! / (n-k)!$ count **ordered** ones.
- **Random variables.** For a discrete X , $E[X] = \sum x \cdot P(x)$. $\text{Var}(X) = E[X^2] - (E[X])^2$.
- **Special distributions** (as lectured).

- Bernoulli(p): mean p , variance $p(1-p)$.
 - Binomial(n, p): mean np , variance $np(1-p)$.
 - Uniform, Poisson, Normal, and Exponential follow if covered.
 - **Covariance and correlation.** $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$. $\text{Corr} = \text{Cov} / (\sigma_X \cdot \sigma_Y)$, which always lies in $[-1, 1]$. Correlation measures **linear** association only. Check which denominator the course uses for sample covariance (n or $n - 1$).
 - **Bayes, LOTP, independence.** These are the published Quiz 04 topics; see the Quiz 04 Practice Guide.
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4. Light practice (self-contained)

- L1 (recall).** Define independence, and write correlation in terms of covariance.
- L2 (combinations).** How many 3-person committees can be chosen from 7 people? What is the probability of exactly 2 heads in 4 fair coin flips?
- L3 (random variable).** For a fair six-sided die X , find $E[X]$ and $\text{Var}(X)$.
- L4 (binomial).** $X \sim \text{Binomial}(10, 0.3)$. Find $E[X]$ and $\text{Var}(X)$.
- L5 (covariance/correlation).** For $X = [1, 2, 3]$ and $Y = [2, 4, 6]$, find the sample covariance (denominator $n - 1$) and the correlation.
- L6 (Bayes).** $P(H) = 0.2$, $P(E | H) = 0.9$, $P(E | \text{not-}H) = 0.3$. Find $P(H | E)$.
- L7 (independence).** $P(A) = 0.25$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.1$. Are A and B independent?
- L8 (visualization).** Pick a chart for (a) the distribution of house prices, (b) price vs. square footage, (c) average price by neighborhood.
- L9 (pandas).** Write one line that gives the mean score for each team in DataFrame `df`.
- L10 (planning).** Why must the tentative Oct 20 date be re-verified before making a dense plan?
- L11 (Content grain — Practice recommendation).** Two labs: Lab A has 90 students (30% prefer Python); Lab B has 60 students (70% prefer Python). (a) $P(\text{prefer Python})$. (b) Given prefer-Python, $P(\text{Lab B})$. (Cite Content quiz01_03 Quiz-03 style; numbers changed.)

4b. Format reminder — Practice recommendation

The Gradescope format notes show that quizzes are about 1.5 points on a single page, and each homework has **two** items (written and Code). The **midterm format is not published**. For practice, write L2–L7 by hand, one per half-page, with every step shown. This is a generic habit, not a claimed midterm format.

5. ANSWER KEY

L1. Independence means $P(A \text{ and } B) = P(A) \cdot P(B)$. $\text{Corr}(X, Y) = \text{Cov}(X, Y) / (\sigma_X \cdot \sigma_Y)$.

L2. $C(7, 3) = 35$. $P(\text{exactly 2 heads in 4 flips}) = C(4, 2)/2^4 = 6/16 = \mathbf{0.375}$.

L3. $E[X] = 3.5$. $E[X^2] = 91/6$. $\text{Var}(X) = 91/6 - 12.25 = \mathbf{35/12 \approx 2.917}$.

L4. $E[X] = 10 \cdot 0.3 = \mathbf{3}$. $\text{Var}(X) = 10 \cdot 0.3 \cdot 0.7 = \mathbf{2.1}$.

L5. The means are 2 and 4. The deviations are (-1, 0, 1) for X and (-2, 0, 2) for Y, so the sum of products is $2 + 0 + 2 = 4$. Sample covariance = $4 / (3 - 1) = \mathbf{2}$. The sample standard deviations are 1 and 2, so correlation = $2 / (1 \cdot 2) = \mathbf{1}$, a perfect positive linear relationship. With the population denominator, the covariance is $4/3$, and the correlation is still 1.

L6. $P(E) = 0.9 \cdot 0.2 + 0.3 \cdot 0.8 = 0.42$. $P(H | E) = 0.18 / 0.42 = \mathbf{3/7 \approx 0.429}$.

L7. $0.25 \times 0.4 = 0.1 = P(A \text{ and } B)$. **Yes, independent.**

L8. (a) Histogram (or box plot). (b) Scatter plot. (c) Bar chart.

L9. `df.groupby("team")["score"].mean()`

L10. The syllabus marks the date as "tentative". The context rule is never to treat an unconfirmed date as fixed, so re-check on D2L, the course site, or in class during Oct 13–19 before planning around it.

L11. (a) $0.3(90/150) + 0.7(60/150) = 0.30.6 + 0.70.4 = 0.18 + 0.28 = \mathbf{0.46}$. (b) $0.28/0.46 = \mathbf{14/23 \approx 0.609}$.

6. Common mistakes + readiness (light)

- Studying regression or neural networks early.
- Treating the tentative Oct 20 date as fixed.
- Skipping Quiz 04 fluency, since Bayes, LOTP, and independence are the only **published** subset.
- Mixing up combinations (unordered) with permutations (ordered).
- Reading a correlation near 0 as "no relationship", when it only means no **linear** relationship.
- Inventing quiz dates from the Tuesday cadence.

Ready (light):

- ☐ Weeks 1–8 concept map reviewed
- ☐ Quiz 04 topics solid
- ☐ L1–L11 correct
- ☐ Date and scope re-verified during Oct 13–19 **before** densifying

7. Prep plan (light — no day-before cram that assumes a confirmed date)

Window	Action
Now – early Oct	Quiz 04 practice and HW3/HW4 probability. Not a midterm marathon.
Sat Oct 3	One light pass over the 380-L- * concept map and L1–L10

Tue Oct 13	HW4 due 11:59 PM (written plus Code)
Oct 13–19	Re-verify the date and scope on D2L, the site, or the tracker. Densify only after that.
If Oct 20 is confirmed	Sit the midterm at 5:00–6:15 PM. The night before, calm review only.

Do deeper work Mon/Wed/Fri and weekends; keep Tuesdays and Thursdays light.

8. Changes in this review (2026-09-24)

- Kept the **date NV** and **LIGHT** banners unchanged in meaning. Changed "PT" to Arizona time.
- **Made the answer key self-contained.** The old key said "see the Quiz 04 guide" for Bayes; all L-items now have worked answers.
- Added the week-3 **Combinations** topic, which was merged into independence and had no practice. Added practice on random variables, the binomial distribution, covariance/correlation, visualization, and pandas (L2–L5, L8–L9).
- Noted that the weeks 6–8 lectures have not happened yet.
- Kept the new practice to ten short items so the guide stays light.

8b. Content-fold (2026-09-24, America/Phoenix)

- Light only: cited Content quiz01_03 + HW01-03; added L11 similar-style LOTP+Bayes drill. Date/scope banners unchanged (NV / PROVISIONAL / LIGHT). No dense invent.

9. Disclaimer

Study aid only — not for submitted work. Follow CSC 380 and UA policies. The title marks provisional/NV status on purpose.